WORKED SOLUTION

STEP I 2003, Question 2

Pure • Algebraic Identities

STEP I 2003, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s1-q2

The key idea

The candidate's answer is right exactly when it satisfies the equation, so substitute it and clear the fractions. Each part then turns into a polynomial condition on the given numbers. In the first part that condition is a quadratic form that can't vanish for real non-zero numbers. In the second it's a cubic expression that factorises beautifully, and the factors are exactly the three equations we're asked about.

Solution

The first question

Throughout, a and b are real and non-zero. The candidate's answer is correct precisely when $x = a + b$ satisfies the equation. For that to make sense at all we need $a + b \neq 0$, and then we need

$$\frac{1}{a+b} = \frac{1}{a} + \frac{1}{b} = \frac{a+b}{ab}.$$

Multiplying through by $ab(a+b)$, which is non-zero, this is equivalent to

$$ab = (a+b)^2, \qquad \text{that is,} \qquad a^2 + ab + b^2 = 0.$$

Completing the square,

$$a^2 + ab + b^2 = \left(a + \tfrac{1}{2}b\right)^2 + \tfrac{3}{4}b^2.$$

Both terms are squares of real numbers, so both are ≥ 0, and the second is strictly positive because $b \neq 0$. Hence $a^2 + ab + b^2 > 0$ for every real a and non-zero real b, so the condition can never hold.

(If $a + b = 0$, the candidate's "answer" $x = 0$ doesn't even make sense in the equation, so that case is no better.) Therefore there are no values of a and b for which $x = a + b$ is correct, as required.

The second question

Now a, b, c are non-zero. The answer $x = a+b+c$ is correct if and only if $a+b+c \neq 0$ and

$$\frac{1}{a+b+c} = \frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{bc+ca+ab}{abc}.$$

Multiplying through by $abc(a+b+c) \neq 0$, this is equivalent to

$$(a+b+c)(ab+bc+ca) - abc = 0. \qquad (*)$$

Ask yourself How can we spot that the left-hand side of $(*)$ factorises?

The target result tells us to expect the factors $a+b$, $b+c$ and $c+a$. Test one: put $b = -a$. Then $a+b+c = c$ and $ab+bc+ca = -a^2 - ac + ca = -a^2$, so the left-hand side becomes $c(-a^2) - (-a^2 c) = 0$. So $a+b$ is a factor, and by symmetry so are $b+c$ and $c+a$. The left-hand side has degree 3, so it is a constant multiple of $(a+b)(b+c)(c+a)$; comparing the coefficients of $a^2 b$ (it's 1 on both sides) shows the constant is 1. Alternatively, expand both:

$$\begin{aligned}(a+b+c)(ab+bc+ca) - abc &= a^2b + a^2c + b^2a + b^2c + c^2a + c^2b + 3abc - abc \\ &= a^2b + a^2c + b^2a + b^2c + c^2a + c^2b + 2abc \\ &= (a+b)(b+c)(c+a).\end{aligned}$$

So $(*)$ is the same as

$$(a+b)(b+c)(c+a) = 0.$$

Now we prove both directions.

If the answer is correct. Then $a+b+c \neq 0$ and $(*)$ holds, so $(a+b)(b+c)(c+a) = 0$. A product of numbers is zero only if one of the factors is zero, so at least one of $a+b=0$, $b+c=0$, $c+a=0$ holds.

Conversely, suppose at least one of these holds; by symmetry say $a+b=0$. Then $a+b+c = c \neq 0$, so the candidate's answer is a legitimate non-zero number, and the factorised form of $(*)$ holds. Since the steps from the equation to $(*)$ were reversible (we only multiplied by the non-zero number $abc(a+b+c)$), $x = a+b+c$ satisfies the equation. You can also see this directly: $\frac{1}{a} + \frac{1}{b} = \frac{1}{a} - \frac{1}{a} = 0$, so the equation reads $\frac{1}{x} = \frac{1}{c}$, whose solution is $x = c = a+b+c$.

Hence the candidate's answer is correct if and only if at least one of $a+b=0$, $b+c=0$, $c+a=0$ holds. ■

Notice that the argument never used that a, b, c are real; the factorisation is an algebraic identity, so the result holds for non-zero complex numbers too. The first part, by contrast, really does need a and b real: over the complex numbers, $a^2 + ab + b^2 = 0$ has solutions such as $b = a\,\mathrm{e}^{2\pi\mathrm{i}/3}$.

Checks

- Take $a = 2$, $b = -2$, $c = 5$: then $\frac{1}{2} - \frac{1}{2} + \frac{1}{5} = \frac{1}{5}$ and $a + b + c = 5$, so the candidate is right.
- Take $a = b = c = 1$: the true answer is $x = \frac{1}{3}$, but the candidate says 3; and indeed $(a+b)(b+c)(c+a) = 8 \neq 0$.
- Expand $(a+b)(b+c)(c+a)$ at $a = 1$, $b = 2$, $c = 3$: it gives $3 \cdot 5 \cdot 4 = 60$, while $(a+b+c)(ab+bc+ca) - abc = 6 \cdot 11 - 6 = 60$.
- We tested both directions exhaustively on all 17 576 triples of non-zero rationals n/d with $|n| \leq 6$ and $d \in \{1, 2, 3\}$, using exact fractions, and found no exceptions.

Common slips

- Writing $a^2 + ab + b^2 = (a+b)^2 - ab$ and stopping: that doesn't show positivity. Completing the square does.
- Proving only one direction in the second part. "If and only if" needs both, and the converse must check that $a + b + c \neq 0$.
- Forgetting that $abc(a+b+c)$ must be non-zero before multiplying through, which is what makes the steps reversible.

Beyond STEP

The expression $(a+b)(b+c)(c+a)$ is a symmetric polynomial, and the identity $(a+b)(b+c)(c+a) = e_1e_2 - e_3$, where $e_1 = a+b+c$, $e_2 = ab+bc+ca$ and $e_3 = abc$, is an instance of the fundamental theorem of symmetric polynomials: every symmetric polynomial is a polynomial in these elementary ones. The same e_1, e_2, e_3 are, up to sign, the coefficients of the cubic with roots a, b, c, which is why they turn up throughout the theory of equations. Allenby's *Rings, Fields and Groups* discusses polynomial rings in this spirit, and Siklos's *Advanced Problems in Mathematics* has more STEP practice in spotting factorisations.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.